HOW BRAINS AND MACHINES LEARN FROM EACH OTHER

A month of daily reading and writing tracing neuroscience's influence on AI, AI's return influence on neuroscience, and where both land in clinical care and in nursing & midwifery education.

Week 1: What early AI actually borrowed from the brain

The emergence of **Artificial intelligence (AI)** as it is known today has long been developed on the backbone of **neuroscience** and **psychology**. Neuroscience is the scientific study of the nervous system and the brain, including their structure, function, development and disorders. While neuroscience studies the physical brain, psychology looks at the mental state and lived experiences. Artificial Intelligence is the capability of a computer system or machine to perform tasks that normally require human intelligence, such as learning, reasoning, and problem-solving. Over the past decade, there has been enormous progress in the area of artificial intelligence.

Going into this, I assumed AI and neuroscience were always somewhat separate fields that occasionally borrowed from each other. What surprised me is how tangled up they used to be; in the early days of computing, work on artificial intelligence and work on the brain weren't really separate projects at all. It's only in the last couple of decades, as both fields grew more complex and specialized, that they drifted apart into their own silos.

The clearest example of the borrowing is **convolutional neural networks (CNNs)**. These weren't designed in a vacuum, they came directly out of Hubel and Wiesel's Nobel-winning work recording from cells in the cat visual cortex, which showed how the brain filters and pools visual information in stages. Features like normalization and pooling that show up in modern CNNs are essentially engineered versions of what those neurons were already doing.

Reinforcement learning (RL) has a similar backstory. It wasn't invented purely as a math problem but rather, it grew out of real observations about how animals learn from reward, and later work tied it explicitly to dopamine activity in the brain. Even things like "**attention**" in modern architectures and "**experience replay**" in RL systems trace back to ideas about how memory and attention work biologically; replay in particular was inspired by theories of how the hippocampus and other memory systems interact.

What stood out to me is that none of this was just loose metaphor. They were specific mechanisms taken from real biological findings and turned into working algorithms. And yet, even with that lineage, current AI still struggles with things the human (mammalian) brain handles almost effortlessly, like energy efficiency (***a brain runs on ~20 watts while training something like GPT-3 takes over 1000 megawatt-hours***). This gap alone made me want to keep reading with the question: **if the borrowing worked so well before, why did the fields stop talking to each other?**

Week 2: Where the AI-brain analogy holds, and where it clearly doesn't

The limitations of current AI have been organized into three categories:

- the inability to interact with the physical world,
- a mode of learning that is both inflexible and fragile, and
- an inefficiency in energy and data that limits who can build AI and where it can operate.

This stretch pushed me past "***AI copied the brain in a few places***" into something more specific i.e. which parts of the analogy are doing real work, and which parts are just convenient framing. The CNN story kept coming back; Hubel and Wiesel's visual cortex work, now four decades old, is still cited as the direct ancestor of modern vision systems. Same with the perceptron, it didn't just take loose inspiration from neurons, it established that

learning happens at the level of synaptic connections, which is still the basic assumption every neural network runs on today. Even **dropout**, where individual neurons in a network are randomly deactivated during training to prevent overfitting, draws inspiration from the brain's own noisy, occasionally misfiring neurons. What struck me is that this wasn't an accidental resemblance, it was engineers looking at a biological quirk and asking what it was actually good for, then building it in on purpose.

But the more interesting part of these days was seeing where the analogy ran out. Current AI is trained passively on fixed datasets; text, images, videos someone else collected. A child learning the word "heavy" spends months physically picking things up before the word even attaches to anything; the meaning is anchored in touch and proprioception, not co-occurrence statistics. A human fingertip alone has around ***240 mechanoreceptors per square centimetre***, a level of grounded, embodied sensing nothing in current AI comes close to. That's a big part of why AI still struggles once it leaves the page; it has no comparable "***I don't know what I don't know***" signal the way a person who has never done plumbing simply knows they can't fix a pipe. It hallucinates instead of flagging uncertainty, because it has no episodic memory of its own experience to check itself against.

The other gap that stuck with me is that AI models are frozen the moment training ends. A midwife gets better with every client she attends to; a self-driving car that hits a weird intersection today will make the exact same mistake tomorrow unless someone manually retrains it. That's a fundamentally different relationship with experience than what human brains have.

By the end of the week, all of this started to converge on a single idea that the various journals kept circling without quite naming until now: **NeuroAI**. On one hand, AI aims to replicate intelligent behaviour, and the most direct path to that goal is reverse-engineering the

brain. On the other hand, neural networks are the closest model yet of distributed, brain-like computation, one uniquely capable of solving complex problems. As one of the papers put it, borrowing Richard Feynman's famous dictum, "***That which I cannot create, I do not understand***," building artificial versions of brain function is itself a way of testing how much we really understand the brain.

What ties both halves of the week together is that this relationship was never one-directional charity from neuroscience to AI. As one journal put it plainly, "***it is important to note that the relationship between AI and neuroscience is truly mutualistic and not parasitic***." Von Neumann's own 1945 report on his computer architecture devoted a whole chapter to how brain-like the design was, citing only McCulloch and Pitts' original neuron paper. And now, going forward, better AI models are actually used as models of the brain itself, feeding insight back the other way. So, I stopped thinking of it as AI copying biology and moving on. It felt more like two fields checking their homework against each other, and heading into Week 2, that's exactly the direction I want to follow: not what neuroscience gave AI, but what AI is starting to give back.

REFERENCES

Hassabis, D., Kumaran, D., Summerfield, C., & Botvinick, M. (2017). Neuroscience-inspired artificial intelligence. *Neuron*, *95*(2), 245-258.

Sadeh, S., Clopath, C. The emergence of NeuroAI: bridging neuroscience and artificial intelligence. *Nat. Rev. Neurosci.* **26**, 583–584 (2025). https://doi.org/10.1038/s41583-025-00954-x

Zador, A., Escola, S., Richards, B. *et al.* Catalyzing next-generation Artificial Intelligence through NeuroAI. *Nat Commun* **14**, 1597 (2023). https://doi.org/10.1038/s41467-023-37180-x

Zador, A., Fellous, J. M., Sejnowski, T., Adam, G., Aimone, J. B., Akwaboah, A., ... & Zucker, S. W. (2026). Neuroai and beyond: Bridging between advances in neuroscience and artificialintelligence. *arXiv preprint arXiv:2604.18637*.

Zador, A. (2024) NeuroAI: A field born from the symbiosis between neuroscience, AI https://doi.org/10.53053/EOIW4593